

1.1. An autonomous vehicle

Input: Information gathered from multiple sensors (such as cameras and GPS) that monitor the environment in real-time.

Output: Control commands for the vehicle's mechanisms (acceleration and signalling).

Justification: An autonomous vehicle requires ongoing data about its surroundings to make informed decisions. The inputs consist of raw data utilized by the machine learning framework to interpret the environment, while the outputs represent the immediate actions taken by the system to navigate the vehicle safely, effectively substituting human operation.

1.2. Netflix's content recommendation engine

Input: The viewing history of the user, ratings, search queries, demographic details, and the viewing patterns of comparable users.

Output: A customized, prioritized list of recommended films or television series.

Justification: The objective of the system is to propose pertinent content to a user from an extensive library. The inputs offer insights into user preferences and behaviours, which the machine learning algorithm leverages to forecast items the user is likely to appreciate, presenting these forecasts as the output.

1.3. Signature verification

Input: A digital representation of a signature or a series of pen strokes (including timing and pressure data).

Output: A binary classification (for instance, "Genuine" or "Counterfeit") or a confidence score reflecting the probability of authenticity.

Justification: The system aims to authenticate an individual's identity based on their signature. The input comprises the data collected from the signature under examination, while the output is the result of the machine learning model's evaluation of this input against a database of known authentic signatures.

1.4. Clinical diagnosis

Input: Patient information encompassing symptoms, medical history, laboratory results, medical imaging (such as X-rays), and possibly genomic data.

Output: A likely diagnosis, a recommended treatment strategy, or a risk assessment score for a particular condition.

Part 2: Regression or Classification Algorithms

2.1. Classifying emails as either promotional or social based on their content and metadata.

Algorithm: Classification

Justification: This task entails categorizing data into specific, predefined classes or labels ("promotional" or "social"). Classification algorithms are specifically designed for such discrete output predictions.

2.2. Forecasting a company's stock price using historical data and market trends.

Algorithm: Regression

Justification: The prediction of a stock price requires forecasting a continuous numerical value. Regression algorithms are employed when the output variable is a real or continuous value.

2.3. Categorizing images of animals into various species based on their visual characteristics.

Algorithm: Classification

Justification: The objective is to assign each image to one of several distinct, predefined categories (animal species). This represents a classic multi-class classification problem.

2.4. Estimating the probability of a patient having a specific disease based on their medical history and diagnostic test results.

Algorithm: Classification

Justification: While this involves predicting a "likelihood" (which is numerical), the primary outcome is generally a decision regarding the presence of a disease (positive/negative, often represented as a binary class). Classification models (particularly logistic regression or similar classifiers) are typically utilized to produce a probability score that is subsequently mapped to a class.

3.1. Identifying anomalies in a manufacturing process through sensor data without prior knowledge of specific anomaly patterns: Unsupervised learning is suitable.

Justification: Unsupervised algorithms are employed with unlabelled data to uncover hidden structures and patterns. Given the absence of prior knowledge regarding specific anomaly patterns (indicating that there are no labelled examples of anomalies available for model training), an unsupervised approach can detect observations that significantly diverge from the established "normal" behaviour based on the inherent structure of the data.

3.2. Estimating customer lifetime value using historical transaction data and customer demographics: Supervised learning (specifically, regression) is suitable.

Justification: This represents a prediction problem involving historical data where the target variable (customer lifetime value) is a continuous numerical value. A supervised model can be trained on a labelled dataset (previous customer data with known lifetime values) to understand

the relationship between input features (transaction data, demographics) and the target value, facilitating predictions for new customers.

3.3. Classifying customer demographics based on their purchase history, browsing behaviour, and preferences: Unsupervised learning (specifically, clustering) is suitable.

Justification: The objective here is to categorize customers into distinct segments or clusters based on similarities in their behaviour and demographics without predefined labels for these groups. Unsupervised learning algorithms, such as clustering, are particularly effective at revealing these natural groupings and hidden structures within unlabelled data.

3.4. Examining social media posts to categorize them into various themes: Supervised learning (specifically, classification) is suitable if a predefined set of themes and labelled training data are available.

Justification: In this context, there would typically be a set of predefined categories (themes) and a labelled dataset of social media posts that have already been assigned to those themes.

4.1. Forecasting fraudulent financial transactions utilizing a dataset predominantly classified as either fraudulent or legitimate.

Inappropriate.

Justification: Semi-supervised learning proves most beneficial when there exists a substantial amount of unlabelled data alongside a limited quantity of labelled data. In this instance, the dataset comprises mostly labelled transactions, rendering it more suitable for conventional supervised learning algorithms, which typically yield greater accuracy and interpretability when ample labelled data is present. Additionally, real-world fraud datasets often exhibit a considerable class imbalance (with a scarcity of fraudulent cases relative to legitimate ones), a challenge that is frequently tackled through resampling techniques within a supervised learning context.

4.2. Evaluating customer satisfaction surveys where merely a small fraction of the data is annotated with satisfaction ratings.

Appropriate.

Justification: This situation exemplifies optimal conditions for semi-supervised learning. Annotating a vast number of customer surveys (for instance, through manual sentiment analysis) is labour-intensive and costly. Semi-supervised algorithms can exploit the extensive amount of available unlabelled data to discern general data structures and patterns, employing the limited labelled data to speed up the classification process, thus enhancing model performance while minimizing manual input.

4.3. Detecting spam emails within a dataset where most emails are categorically labelled.

Inappropriate.

Justification: Analogous to the initial scenario, the presence of a majority of emails already labelled (as spam or not spam) indicates that there is adequate labelled data to construct a robust model using standard supervised learning methodologies. In this context, semi-supervised learning would not confer substantial advantages over supervised approaches, as the primary issue of data scarcity is absent.

4.4. Estimating the likelihood of default for credit card applicants based on their comprehensive financial and credit-related data.

Inappropriate.

Financial institutions generally possess extensive historical data regarding applicants who have defaulted as well as those who have not. This dataset is entirely labelled and is utilized to train models (such as logistic regression, random forests, or gradient boosting) to evaluate the risk associated with new applicants. Given the availability of comprehensive and labelled historical data, a supervised approach is the conventional and most suitable method.